

Hybrid Biomedical Nuclei Image Analysis

Github Link : <https://github.com/Pampapathi25/AI-imaging-Code-And-Report>

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1. Project aim and experimental workflow

The goal was to develop and test a biomedicine image analysis flow where generative AI was complemented by quantifiable image evidence. The pipeline looked like: raw image -> standardised grayscale image -> description via a direct VLM -> classical/Otsu analysis -> segmentation via the U-Net -> region semantics-level quantitative features -> structured JSON -> short narrative. It is essentially a solution for one of the main problems that hinders the progression of medical AI: a fluent language-model response is not automatically reliable. Each and every narrative that is generated can be related back to quantitative evidence via retaining masks, feature tables, and records in JSON structure.

The dataset provided for nuclei segmentation comprised 120 images with nuclei, including 80 training images, 20 images (held out) to validate the performance, and 12 test (unseen) images. All the images were displayed at 256x256 resolution. Pixel-level ground truth for segmentation was provided by training/validation masks. There was intention to be explicitly educational, not to be a diagnosis with any output.

The original brief called for the use of llama3.2-vision. This crashed with a familiar 'unknown model architecture: mllama' error when running locally on our Mac machine. Arou was used with the local vision model of Qwen2.5-VL (qwen2.5vl:3b), in accordance with the written permission of the instructor. Text-only, numbers-first interpretation and final narratives were performed using the Qwen2.5:3b model. This maintained the original LLM comparison in the local area, but kept the analysis objectives separation unchanged.

2. Data preparation, EDA and direct VLM description

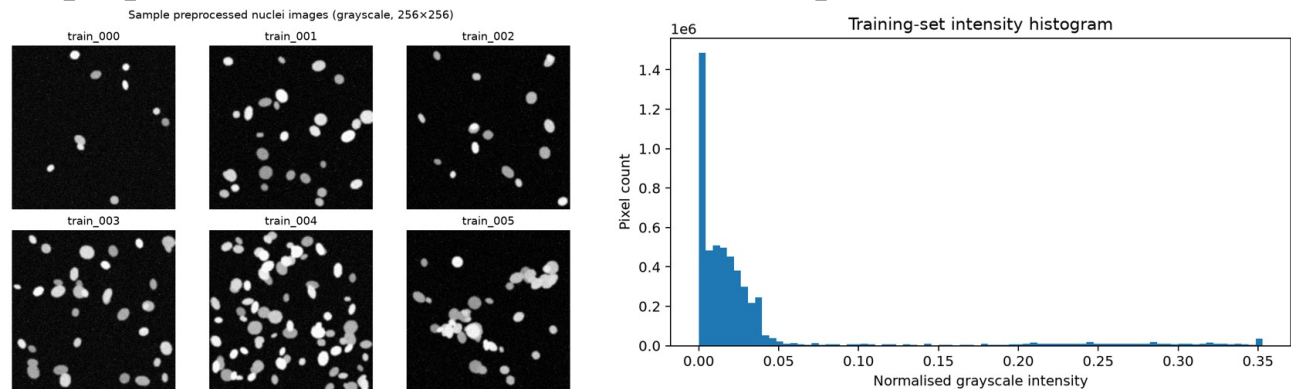


Figure 1. Representative prepared nuclei images/masks (left) and the training-set grayscale intensity distribution (right).

Bright, compact nuclei-like visible fields (shown by the EDA) fill the diagram more or less densely on a dark background. There is a strong central peak, with a small bright tail of contribution, as can be seen in this histogram. This separation is crucial because it provides a potentially realistic baseline, however there are variations in the size of the objects, levels of brightness between the objects, degree of crowding between objects and proximity to the objects. Such variations inspire a learned segmentation model that does not rely on a global threshold but instead is guided by the local spatial context.

Prompt engineering and repeatability

The result of a naive prompt to 'Describe this biomedical image' was a serious error - the VLM although showing my fluorescence image interpreted it as an X-ray/radiograph and invented bones/Odysseus body parts. Such a compact structure for the JSON output (modality, tissue_type, notable_features, image_quality) is imposed on the model by the optimized prompt, which also needs to return 'uncertain'. This optimized prompt also anchors the model to the known educational context, disallows diagnosis and clinical significance, and requires that the model output be a compact JSON structure with fields for modality, tissue_type, notable_features and image_quality, where the latter field must be set to 'uncertain'.

When the images were scanned three times using optimised acquisition settings, these two timeseries both identified fluorescence microscopy and the blue fluorescent spots. Of the 2 that returned tissue_type='uncertain' and one returned 'Nuclei' This illustrates that there are significant improvements, within the relevance and format, if the constraints are enforced in addition to the feature but no amount of such timing constraints can render the feature completely deterministic. Replication was then useful as a useful indicator of stability.

3. Classical segmentation, region features and numbers-first LLM

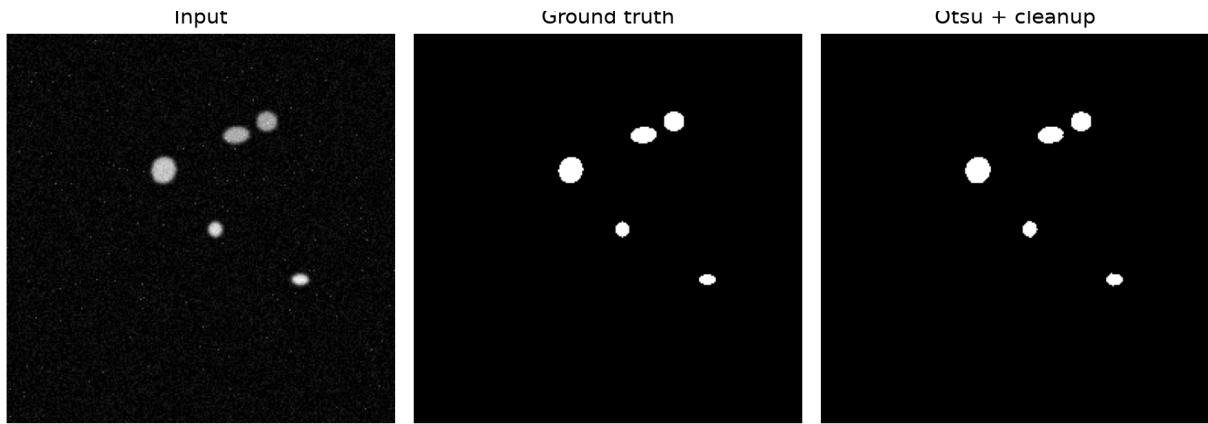


Figure 2. Classical pipeline example: grayscale input, cleaned Otsu foreground and supplied reference mask.

Separation of measurement and interpretation was a conscious aim of the classical approach. The data-driven global intensity thresholding by Otsu, the morphological cleanup of isolated noise, and the connected-component labelling process were carried out to separate different object areas, while regionprops_table obtained meaningful object features. Presented object area, eccentricity, solidity and mean intensity, and with aggregate measures being object count and occupied area fraction.

Classical pipeline detected the classical pipeline detected the 5 objects on the val_000 with mean area of 131.4 pixel, area fraction of 0.010, mean eccentricity of 0.519, mean solidity of 0.955 and mean object intensity of 0.224. The rule-based summary determined the field to be sparse and the field shapes mostly regular. The content was extracted from the image, converted to short numerical summary, and sent to text LLM with the shortcuts of which the model was never exposed to the image. The prompt has an explicit ban on diagnosis, included imaginary details and unsupported conclusions, and asked for JSON fields n_objects, density_class, shape_regularity and quality_flag.

This approach, which used numbers first, was much more reliable for quantitative claims, in that each claim was able to be tested against the feature table. However, one output specified key 'rule_based_class' rather than the desired key 'density_class'. That isn't a cosmetic problem, that's an important result: an appropriate grounding does not imply schema compliance. In a stronger system, type/range checks and the determination of the record should provide repair/rejection, with the addition of JSON-schema validation before accepting an LLM record.

4. U-Net training and quantitative evaluation

Method	Mean Dice	Mean IoU	Interpretation
U-Net	0.9962	0.9924	Near-complete mask overlap
Otsu	0.9788	0.9584	Strong deterministic baseline

Table 1. Mean segmentation performance on the 20-image held-out validation set.

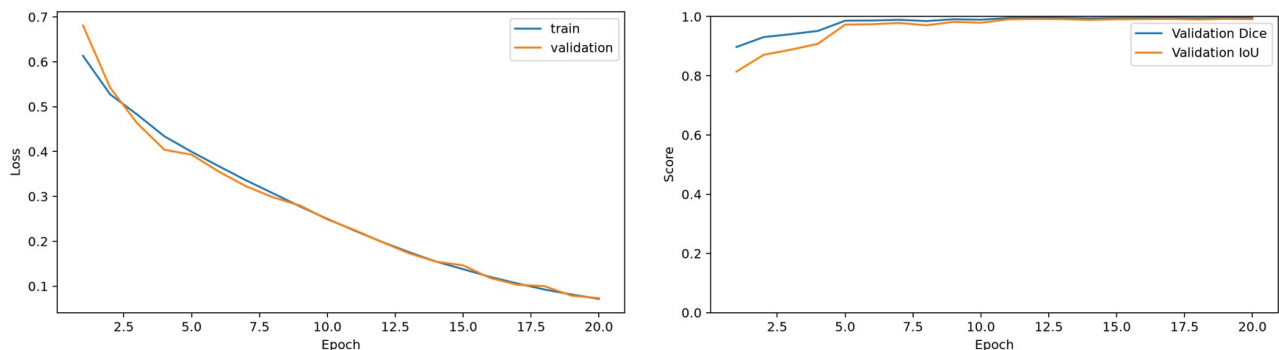


Figure 3. Training/validation loss (left) and validation Dice/IoU (right) across 20 epochs.

The small U-Net was able to train the segmentation task in a short number of iterations. Validation Dice improved from epoch 1 (0.8975) to epoch 19 (0.9962), and validation IoU had a similar trend of getting better with each epoch (0.8144 to 0.9924 at epoch 19). Training loss decreased from 0.6133 to 0.0716 and validation loss from 0.6809 to 0.0733 by epoch 20. The metrics levelled off around 1.0 for the most part from the later epochs, the biggest gain coming from the early epochs.

Training and validation losses on the included dataset behave similarly and all the validation measures after convergence do not exhibit a large amount of variance, indicating that in the following validation we are not seeing severe overfitting. This does not mean the very high score — it must be considered in its context, however; the sample size was small, and the dataset was synthetic and consistent in appearance. The process of internal validation thus reflects learning that has occurred on this distribution and not generalisation to the clinical context.

5. U-Net versus Otsu: visual and per-image analysis

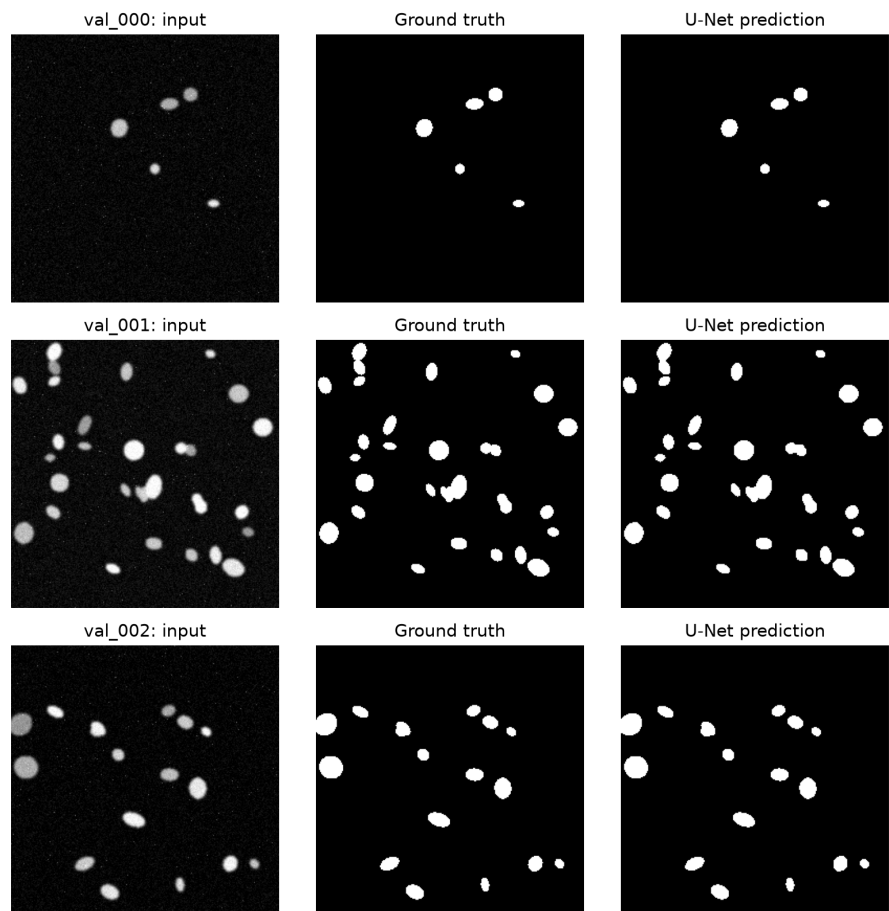


Figure 4. Three held-out examples showing input, ground-truth mask and U-Net prediction.

As seen in the prediction panels, the U-Net is able to closely capture both the boundaries of the isolated and crowded nuclei. In terms of its performance it was better than Otsu quantitatively in all 20 numbers of validation images for Dice and IoU respectively. The largest Dice advantage was val_018 (U-Net 0.9975 vs Otsu 0.9754), followed by val_019 (0.9979 vs 0.9773). The brief requests that you provide an example of the times that each way has worked better: with no image in the validated result recorded having a higher Otsu value. Instead of creating such a case, val_011 is reported as the Otsu result that is closest to the Otsu result; (0.9835 vs U-Net 0.9962).

Dice is more general than IoU, as it considers the region as a whole, as opposed to just the intersection (IoU measures what is intersection divided by the union). Scores close to 1 imply that there is a small percentage of mismatch pixels. The formers are still almost perfectly matched at about 0.9941 Dice, and the other errors are most likely very minor shifts of the boundaries and very few errant foreground pixels and/or embellishments of the area, not real object failures. Otsu is still useful, since it obtained 0.9788 mean Dice without labelled training and optimisation or stochastic inference.

6. Hybrid pipeline on unseen test images

The final stage implemented the concept ‘numbers first, language second’. For each unseen image, the specially trained U-Net segmented it and was asked to calculate connected regions' areas, which were then used to form an abbreviated record of the image's properties (features): image_id, n_objects, mean area, density class and (n object density) quality flag, finally the model was asked to provide a narrative of the image in one paragraph, without the diagnosis task. All of the 12 per-image records were organized into hybrid_test_records.csv for both image and dataset level analysis.

Image	Objects	Mean area	Density	Quality
test_000	8	191.88	sparse	uncertain

test_004	42	337.02	dense	uncertain
test_005	15	434.67	normal	uncertain
test_010	37	354.27	normal	uncertain

Table 2. Selected structured outputs from the unseen test set.

This showed another cryptic oversight: test_005's density_class was normal, but the story's density had been declared to be 'relatively dense'. The text is not all that bad, it just exemplifies that a structured classification will not defeat the intended output by natural language. A well-developed implementation should also be able to match and contrast narrative claims and the JSON; if exact consistency is needed, it should also be able to turn the JSON back into prose.

7. Answer to the Questions

1. Which description is more useful, and which is more trustworthy, the direct VLM description or the numbers-first description and why? The direct VLM can be useful in very broad visual description, for example; it can include characteristics that are not specifically designed as features. The numbers-first approach is more reliable for conclusions like numbers, density and shape, since the measurement of the input is reproducible, and it can be inspected. The naive VLM's X-ray hallucination shows how visual alone is not enough. The most desirable design thus is to be able to use the optimised VLM as a supplement to description and the numerical record for factual claims.

2. Did the U-Net improve on classical Otsu segmentation for your modality? Give one example image where each approach did better. Yes. Mean Dice improved from 0.9788 to 0.9962 and mean IoU from 0.9584 to 0.9924. val_018 gives a clear U-Net advantage. When the val_011 becomes close, it means that the thr is close to the best result when the foreground/background contrast is favourable; but Otsu never out does U-Net on this validation set.

3. Report your U-Net's Dice and IoU. What do these numbers mean, and where does the model tend to make its mistakes (which images or regions)? The spatial agreement is very high for both metrics with the provided masks - the stricter being the IoU. The errors are focused on the fine object boundaries and small discrepancies in the foreground and not large missed boundaries. Performance metrics on microscopy that has a different level of noise (or other acquisition methods) should not be extrapolated from these scores, as the images are synthetic.

4. Where in the pipeline can the LLM hallucinate, and what design choices reduce that risk? Why does keeping the structured JSON as the "source of truth" help? There are three significant opportunities. First, the direct VLM can fail to recognize the modality, the tissue or visible structures as demonstrated by the naive Xray/bone response. Second, the numbers-first LLM might change numbers – or create interpretation, or break the requested schema, like the unsolicited JSON key. Third, the final narrative may also be semantically detached from the recorded narrative, with the 'normal' vs. 'relatively dense' in the final line reflecting the differences. These failures arise from otherwise good image analysis and we should therefore not use language generation as the source of analysis.

Descriptive-not-diagnostic prompts, known-modality context, numbers-only inputs (when the classical interpretation of JSON input is desired), and fixed JSON fields were available as risk-reduction decisions. Preservation of the raw feature tables and including structured JSON alongside all narrative text was also kept as a risk-reduction decision. The next engineering step would be schema validation and cross-checking of counts and/or classes, to make sure they match before showing prose. Using the JSON as the source of truth allows for the review to follow a statement back to feature and mask to make the pipeline auditable, rather than relying on free text that is hard to interpret by an outsider.

5. Considering accuracy, auditability, and the limits of your dataset, would you trust any part of this system in a real clinical setting? What single change would most improve trustworthiness? Not autonomously. The segmentation model has high performance on the provided validation split, but it is small size, synthetic data, and has not been validated in an external study, across labs, scanners, staining protocols and across patient populations. Hallucinations and compliance problems are also detected on the language stages. The biggest impact would be to have external, expert annotation on a very large scale in a representative dataset for the intended clinical modality/acquisition conditions. This would see if observed Dice/IoU survive realistic domain shift or not.

8. conclusion

Overall, from the experiment, it can be concluded that a hybrid architecture is a viable alternative. Otsu offered a solid as baseline, clear and cheap output; U-Net showed best possible segmentation accuracy (Dice: 0.9962, IoU: 0.9924); determinative region features turned masks into readable measurements; and local LLM helped with readability of the output. Most importantly, the project had no notion of language generation as ground truth. Each of these examples demonstrates the naïve VLM hallucination, schema-key deviation and the narrative-density inconsistency, all of which suggests that an apparently plausible response may fail at various extents.

The trade-off is so essentially between some degree of accuracy and descriptive flexibility versus simplicity of audit and deployment. U-Net has a very good accuracy on this dataset, but needs labelled data and could be vulnerable to the domain shift. Otsu is a less accurate, yet deterministic and reproducible method. LLMs enrich communication but with stochasticity engendering potential contradiction. I suggested the following end-to-end architecture: U-Net Segmentation -> Denom Schemas -> JSON (source of truth) -> (optional) LLM Narrative (explanation layer). This way there is no loss of predictability and each downstream statement is inspectable.

References

- Ronneberger, O., Fischer, P. & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI, 234-241.
- Otsu, N. (1979). A Threshold Selection Method from Gray-Level Histograms. IEEE Transactions on Systems, Man, and Cybernetics, 9(1), 62-66.
- van der Walt, S. et al. (2014). scikit-image: Image processing in Python. PeerJ, 2, e453.
- Qwen Team (2025). Qwen2.5-VL model documentation/technical report. Local instructor-approved VLM substitute used in this project.